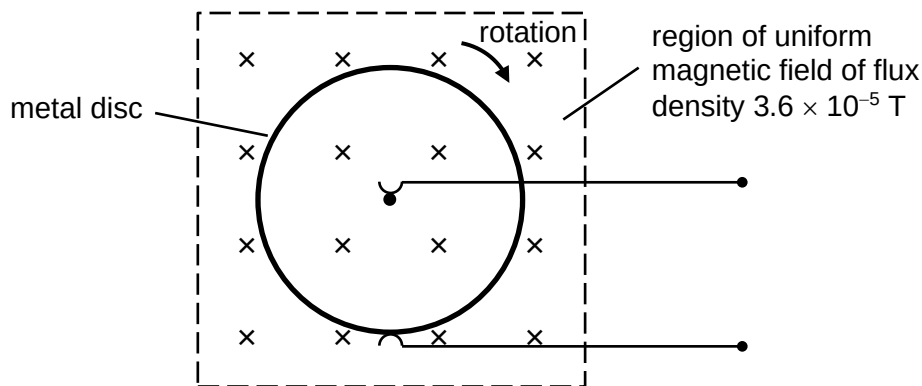


- 26** The diagram below shows a metal disc of area A situated in a uniform magnetic field of flux density $3.6 \times 10^{-5} \text{ T}$. The plane of the metal disc is perpendicular to the magnetic field which is directed into the plane of the diagram. The metal disc is rotated about an axis through its centre at 1500 revolutions per minute. An e.m.f. of $3.7 \times 10^{-7} \text{ V}$ is induced between the centre of the metal disc and its rim.



What is the area A of the metal disc?

- | | | | | | | | |
|----------|----------------------------------|----------|----------------------------------|----------|----------------------------------|----------|-------------------------------|
| A | $2.2 \times 10^{-8} \text{ m}^2$ | B | $4.1 \times 10^{-4} \text{ m}^2$ | C | $3.5 \times 10^{-2} \text{ m}^2$ | D | $5.2 \times 10^4 \text{ m}^2$ |
|----------|----------------------------------|----------|----------------------------------|----------|----------------------------------|----------|-------------------------------|

L2 **Answer: B**

By Faraday's law of electromagnetic induction,

[Turn over

$$E = \left| -\frac{d\Phi}{dt} \right| = \frac{BA}{T} = BAf$$

$$3.7 \times 10^{-7} = (3.6 \times 10^{-5}) A \left(\frac{1500}{60} \right)$$

$$A = 4.1 \times 10^{-4} \text{ m}^2$$